

Tuong Luu

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EDUCATION

University of Waterloo

Candidate for Bachelor of Honours Computational Mathematics

Waterloo, Ontario

Sep. 2023 – Present

QUALIFICATIONS

Languages: Java, Python, C, C++, JavaScript, TypeScript, HTML, CSS

Developer Tools: Git, GitHub, Docker, Linux, Bash

Libraries: Java AWT, Java Swing, React, Express, MongoDB, Node.js

Certifications: [Azure Fundamentals](#), [Azure AI Fundamentals](#)

PROJECTS

Kyzen - 2D Game Engine | Java, LWJGL, Maven, OpenGL

- Developed a 2D game engine with **batch rendering**, optimizing the rendering of multiple objects in a single batch for improved performance
- Utilized **Object-Oriented Programming** (OOP) principles to create scalable software, implementing design patterns like **Builder** and **Singleton** to ensure modularity and maintainability
- Implemented an **Entity-Component System** (ECS), providing a modular and extendable architecture for flexible game object composition
- Integrated **texture atlas** support, enabling **efficient** texture management for sprites and tiles, reducing draw calls

Tetris Clone (Biquadris) | C++, Make, Valgrind, Software Development Lifecycle

- Collaborated in a team of three to design and develop a Tetris game, adhering to the **Software Development Lifecycle** (SDLC) phases, including planning, **UML** modeling, implementation, testing, and deployment
- Utilized Object-Oriented Programming (OOP) principles to create scalable software, implementing design patterns like **Observer**, **Decorator**, and **Factory** to ensure modularity and maintainability
- Performed thorough manual playtesting to verify game mechanics, such as block movement, collision detection, and scoring, identifying and resolving issues to enhance performance and user experience

AniDis - Anime Discussion Platform | React, TypeScript, Node.js, Express, MongoDB, Docker

- Developed a **full-stack** anime discussion platform using the **MERN** stack (**M**ongoDB, **E**xpress, **R**eact, **N**ode)
- Integrated third-party API to fetch **real-time** anime data, including episodes, trending shows, and series details
- Designed and implemented a threaded discussion system, enabling structured conversations and seamless replies between users
- Utilized **Docker** for containerization, improving scalability and deployment flexibility
- Deployed** the application on Render, ensuring accessibility, live at anime-discuss.onrender.com

2D Game | Java, Javadoc, Git

- Designed and developed a top-down 2D game, demonstrating proficient in game development and a deep level of Java's understanding
- Optimized game performance to efficiently handle over **10,000** entities, showcasing expertise in performance tuning within game development
- Applied **Object-Oriented Programming** principles to streamline and enhance entity development, contributing to the efficiency and maintainability of the game codebase

EXPERIENCE

WE Accelerate Program - Azure and Artificial Intelligence

University of Waterloo

Waterloo, Ontario

Jan. 2025 – Present

- Collaborated with a team of 5 to research an AI-based solution to prevent financial fraud against elders
- Designed and implemented a comprehensive planning tool to organize and manage project milestones
- Developed the pipeline sequence wireframe using Microsoft Visio, visualizing each step of the project to ensure a clear and structured workflow
- Worked closely with a project mentor to align project goals with industry standards and best practices